Introducing Shorewall

Linux security is based on the Netfilter system, which is a powerful framework provided by the Linux kernel to perform all types of network operations like packet filtering, NAT, port translation and packet blocking from external sources. Netfilter is implemented via user-space applications and iptables. The latter is powerful, yet complex to work with. So, to make Linux systems more secure, the Shorewall firewall has emerged as a good choice.

Shorewall is a gateway/firewall configuration tool for Linux and is regarded as a high-level tool for configuring Netfilter. All the firewall requirements are entered by users in configuration files. These configuration files are read by Shorewall, and with the support of iptables, iptables-restore, ip and tc utilities, the firewall configures Netfilter in the Linux kernel. Shorewall can be used as a dedicated firewall system or a multi-functional gateway/router/server in the Linux system. Shorewall is a Perl based wrapper for IP tables.

The main objective behind the development of the Shorewall firewall was to create an abstraction in the configuration of the firewall of a higher level, as compared to standard iptables. The advantage of this mechanism is that it divides the interfaces into zones with different levels of access, so that the user can operate on a group of computers, instead of addresses, connected to the interface. In the Shorewall system, users can deploy policies for the zone in an easy and comprehensive manner.

Shorewall is not a daemon running in the background, but is better known as a shell script, which converts configuration files into the iptables commands.

Latest version: 5.2.0.1
Official website: *http:// www.shorewall.org*
Licence: GPLv2
Creator: Thomas M. Eastep

Shorewall firewall configuration files

The following files operate the overall Shorewall firewall:

- */etc/shorewall/shorewall.conf* – This configures global firewall parameters.
- */etc/Shorewall/params* -- This is the file that sets shell variables to expand in other files. It is processed by */bin/sh* or by the shell specified via *SHOREWALL_SHELL* in the */etc/shorewall/Shorewall.conf* file.
- */etc/shorewall/zones* – This partitions the firewall's view of the world into zones.
- */etc/shorewall/policy* – This establishes the firewall's high-level policy.
- */etc/shorewall/initdone* – This is an optional Perl script, which is executed by the Shorewall rules compiler after finalising installation.
- */etc/shorewall/interfaces* – This explains the interfaces on the firewall system.
- */etc/shorewall/hosts* – This file helps users to define zones in terms of individual hosts and sub-networks.
- */etc/shorewall/masq* – This file directs the firewall when to use many-to-one (Dynamic) NAT and Source NAT (SNAT).
- */etc/shorewall/mangle* – This file contains rules for packet marking, TTL, proxies, etc.
- */etc/shorewall/rules* – This file lists exceptional rules to overall policies in */etc/shorewall/policy*.
- */etc/shorewall/nat* – This defines one-to-one NAT rules.
- */etc/shorewall/proxyarp* – This defines rules with regard to proxy ARP.
- */etc/shorewall/tcrules* – This is used for traffic controlling/shaping and policy routing.
- */etc/shorewall/tunnels* – This defines VPN based rules.
- */etc/shorewall/blrules* -- This is for a set of machines that are blacklisted.
- */etc/shorewall/init* – This is for the commands to be executed at the start of Shorewall.
- */etc/shorewall/start* – This is for commands executed after the start of Shorewall.
- */etc/shorewall/stop* – This is for commands executed when Shorewall is stopped.
- */etc/shorewall/accouting* – This is for IP traffic accounting rules.
- */etc/shorewall/providers* – This is for alternate routing tables.
- */etc/shorewall/vardir* – This determines the directory that will maintain the state of Shorewall.

Features of Shorewall

- **Accounting:** This appropriately counts the packets and bytes, using categories and rules specified by network administrators. It is a powerful tool that provides all sorts of information about inbound and outbound traffic.
- **Supports many types of router/firewall applications:** Shorewall is highly efficient in customising all the preferences of users via configuration files, and there are no limitations in network interfaces. It allows administrators to partition the network into zones and provides full administrative control over connections, permitted by every pair of zones.
- **Tunnelling:** Shorewall is efficient in creating tunnels for VPNs, like IPSec, PPTP, GRE, IPIP, OpenVPN, IPv6-over-IPv4, IPv4-over-IPv4 and others.
- **Centralised administration**: It can be monitored and administered via any network connected system. It supports Windows and even Mac OS X.
- **Support for address/routing management:** It is equipped with tons of features which enable masquerading, port forwarding, one-to-one NAT, proxy ARP, NETMAP, multiple ISP support, etc.
- **Support for virtualisation:** Shorewall can efficiently work with a range of virtualisation software like KVM, XEN, Linux-VServer, VirtualBox, LXC and even Docker